Please read the following carefully. ***Lees die volgende noukeurig***

1. Answer all the questions in the space provided on the question paper. *Antwoord alle vrae in die spasie gelaat op die vraestel.*
2. No pencil work will be marked. Do not use "tippex". *Geen potlood-werk sal gemerk word nie. Moet nie uitwis-vloeistof ("tippex") gebruik nie.*
3. If you need more space, please use the back of the pages, and clearly indicate where the question continues. *As u meer spasie benodig gebruik die agterkant van die bladsye. Dui duidelik aan waar die vraag aangaan.*
4. Where applicable you need to draw the necessary diagrams (e.g. free-body diagrams). *Teken diagramme waar nodig (bv. vryliggaams-diagramme).*
5. Your answer needs to be written in a systematic and logical fashion. You must show that you understand and can use the relevant physics. **Only substitute numerical values in the final step of calculations! A correct answer does not imply that you will receive full marks.** *U antwoorde moet sistematies en logies neergeskryf word. U moet bewys dat u die nodige fisika verstaan en kan toepas.* ***Numeriese waardes mag slegs in die laaste stap van 'n berekening vervang word! 'n Korrekte antwoord beteken nie dat u noodwendig vol punte sal kry nie.***
6. **Units must be used when numbers are substituted else marks will be subtracted.** ***Eenhede moet gebruik word wanneer numeriese waardes vervang word, andersins sal punte afgetrek word.***
7. The use of the cheat sheet is only allowed subject to the rules stated on it. *Die formuleblad mag slegs gebruik word onderhewig aan die reëls daarop gedruk.*
8. All exam regulations of the University of Pretoria apply, and any contravention of it will be treated in a serious manner. *Alle eksamen regulasies van die Universiteit van Pretoria geld en enige oortreding daarvan sal in 'n ernstige lig beskou word.*

Constants and formulas / ***Konstantes en formules***
$\varepsilon_0 = 8.854 \times 10^{-12}$ C^2/(N·m^2); $G = 6.674 \times 10^{-11}$ N·m^2/kg^2; $e = 1.602 \times 10^{-19}$ C; $k = 1.381 \times 10^{-23}$ J/K; $m_e = 9.109 \times 10^{-31}$ kg; $m_p = 1.673 \times 10^{-27}$ kg; $R = 8.314$ J/(mol·K); $N_A = 6.02 \times 10^{23}$ /mol; 1 atm = 101.3 kPa; 1 u = $1.6605677 \times 10^{-27}$ kg; $c = 3 \times 10^8$ m/s; $V_{\text{sphere/sfeer}} = \frac{4}{3}\pi r^3$; $F_{gravity} = \frac{Gm_1m_2}{r^2}$; 1 litre = 1000 cm^3; $M_{\text{Earth}} = M_{\text{Aarde}} = 5.97 \times 10^{24}$ kg, ρ(water) = 1 g/cm^3; $r_{\text{Earth}} = r_{\text{Aarde}} = 6.37 \times 10^6$ m

---

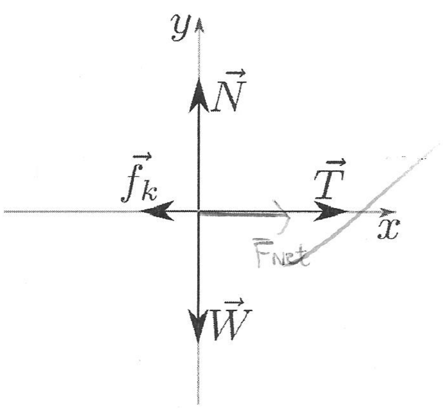
(1) Question 1 / *Vraag 1* [6]

In the following questions, the diagrams show force vectors that are **drawn to scale**. State in each case if the object is in equilibrium or not. Indicate on the figure, the direction of the net force $\vec{F}_{\text{net}}$ (you do not have to draw it to scale!).

In die volgende vrae, wys die diagramme krag-vektore wat volgens ***skaal geteken*** *is. Dui in elke geval aan of die voorwerp in ewewig is of nie. Teken op die figuur, die rigting van die netto krag* $\vec{F}_{\text{net}}$ *(jy hoef dit nie volgens skaal te teken nie).*

(a)

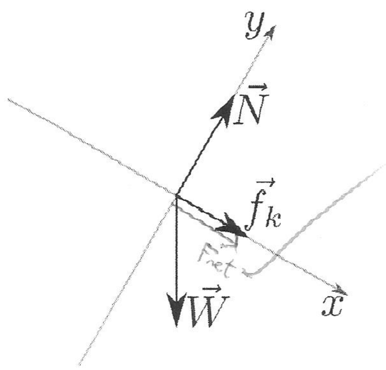
Not in equillibrium.



4 1/2

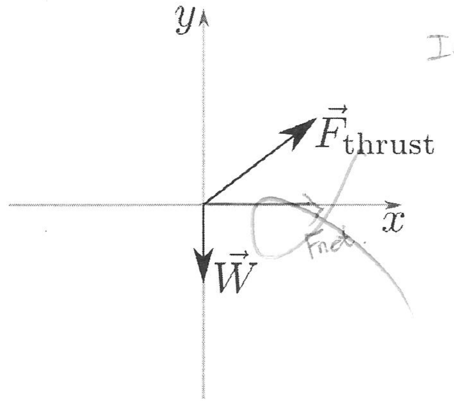
©University of Pretoria / ©*Universiteit van Pretoria*; 2016

(b)



Is not in equillibrium.

(c)

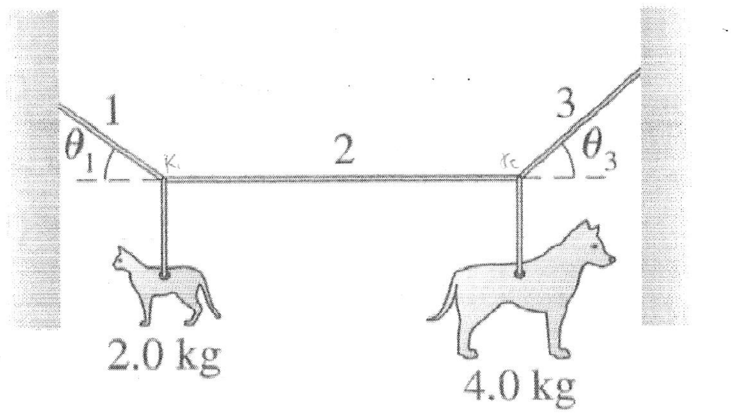


Is not in equillibrium.

2 **Question 2 / *Vraag 2*** [12]

The figure shows an art installation at a museum. It has a 2.0 kg steel cat and a 4.0 kg steel dog that are suspended from a lightweight cable as shown. It is found that $\theta_1 = 20°$ when the center cable (2) is adjusted perfectly **horizontal**. What are the tension and the angle θ_3 in cable number 3?
Use the conventional coordinate system of horizontally to the right to be the positive x-direction, and vertically upwards to be the positive y-direction. **You must use the component method together with the appropriate Newton law and free-body diagrams to solve the problem.**

Die figuur toon 'n kunswerk by 'n museum. Dit bestaan uit 'n 2.0 kg staal kat en 'n 4.0 kg staal hond wat hang vanaf lig-gewig kabels soos aangedui. Daar word gevind dat $\theta_1 = 20°$ wanneer die middel kabel (2) presies ***horisontaal*** *is. Wat is die span-krag en die hoek θ_3 vir kabel 3?*
Gebruik die konvensionele koördinaat-stelsel met na regs en horisontaal die positiewe x-rigting, en vertikaal-opwaarts die positiewe y-rigting. ***Jy moet die komponent-metode gebruik tesame met die nodige Newton wet en vry-liggaam diagramme om die probleem op te los.***



0

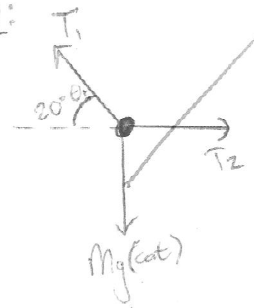
 ©University of Pretoria / © *Universiteit van Pretoria*; 2016

$m_{dog} = 4kg$

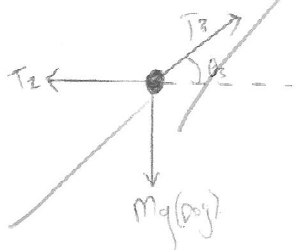
$M_{cat} = 2kg$

$\theta_i = 20°$

Knot 1:



Knot 2:



(12)

12

For knot 2:

x: $\sum \bar{F}_x = ma_x$

$T_3 \cos\theta_3 - T_2 = 0$

$T_3 \cos\theta_3 = T_2$

$T_3 = \frac{T_2}{\cos\theta_3}$ (2)

y: $\sum \bar{F}_y = ma_y$

$T_3 \sin\theta_3 = mg$

$T_3 = \frac{m_{dog}\, g}{\sin\theta_3}$ (3)

For knot 1:

x: $\sum \bar{F}_x = ma_x$

$T_2 - T_1\cos\theta_1 = 0$

$T_2 = T_1\cos\theta_1$ (1)

y: $\sum \bar{F}_y = ma_y$

$T_1 \sin\theta_1 - m_c g = 0$

$T_1 = \frac{m_c g}{\sin\theta_1}$

$= \frac{(2{,}0kg)(9{,}8m/s^2)}{\sin 20}$

$= 57{,}31N$

$\therefore T_2 = T_1\cos\theta_1$ (1)

$T_2 = 57{,}31N \cos 20$

$= 53{,}85N$

$T_3 = \frac{T_2}{\cos\theta_3}$ (2)

$T_3 = \frac{M_{dog}\, g}{\sin\theta_3}$ (3)

Using (2) and (3):

$\frac{T_2}{\cos\theta_3} = \frac{M_{dog}\, g}{\sin\theta_3}$

$T_2 \tan\theta_3 = M_{dog}\, g$

$\tan\theta_3 = \frac{M_{dog}\, g}{T_2}$

$\tan\theta_3 = \frac{(4{,}0kg)(9{,}8m/s^2)}{(53{,}85N)}$

$\theta_3 = 36{,}05°$

$T_3 = \frac{M_{dog}\, g}{\sin\theta_3} = \frac{(4{,}0kg)(9{,}8m/s^2)}{\sin(36{,}05)}$

$= 66{,}61N$ at $36{,}05°$ from positive x axis.

12

©University of Pretoria / ©*Universiteit van Pretoria*; 2016

9

Question 3 / *Vraag 3* [9]

The figure shows the "Atwood machine". Show that the tension in the rope is given by

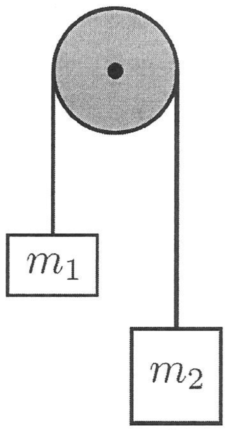
$$T = \frac{2m_1m_2}{m_1+m_2}g$$

if the device is let go of from rest and allowed to accelerate. The mass $m_2 > m_1$.
Use the conventional coordinate system of horizontally to the right to be the positive x-direction, and vertically upwards to be the positive y-direction. **You must use the component method together with the appropriate Newton law and free-body diagrams to solve the problem.**

Die figuur toon 'n "Atwood-masjien". Bewys dat die span-krag in die tou word gegee deur

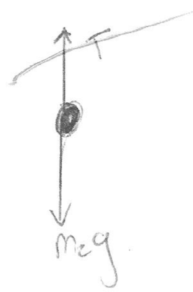
$$T = \frac{2m_1m_2}{m_1+m_2}g$$

as die stelsel losgelaat word uit rus, en toegelaat word om te versnel. Die massa $m_2 > m_1$.
Gebruik die konvensionele koördinaat-stelsel met na regs en horisontaal die positiewe x-rigting, en vertikaal-opwaarts die positiewe y-rigting. ***Jy moet die komponent-metode gebruik tesame met die nodige Newton wet en vry-liggaam diagramme om die probleem op te los.***



For m_1:

For m_2:



m_1:

Y axis: $\sum F_y = m_1 a_y$

$T - m_1 g = m_1 a$

$T = m_1 a - m_1 g$ (1)

m_2:

Y axis: $\sum F_y = m_1 a$

$m_2 g - T = m_2 a$

$T = m_2 g - m_2 a$ (2)

9

 ©University of Pretoria / ©*Universiteit van Pretoria*; 2016

$T = m_1 a - m_1 g$ | $T = m_2 g - m_2 a$

$T - m_2 g = -m_2 a$

$m_2 a = m_2 g - T$

$a = \frac{m_2 g - T}{m_2}$ — (3)

$T = m_1\left(\frac{m_2 g - T}{m_2}\right) - m_1 g$

Subs (3) for a

$= \frac{m_1 m_2 g - m_1 T}{m_2} - m_1 g$

$m_2 T = \frac{m_1 m_2 g - m_1 T}{m_2} - \frac{m_2 m_1 g}{m_2}$

$m_2 T = m_1 m_2 g - m_1 T - m_2 m_1 g$

$m_2 T = 2 m_1 m_2 g - m_1 T$

$m_2 T + m_1 T = 2 m_1 m_2 g$

$T(m_2 + m_1) = 2 m_1 m_2 g$

$T = \frac{2 m_1 m_2 g}{m_2 + m_1}$

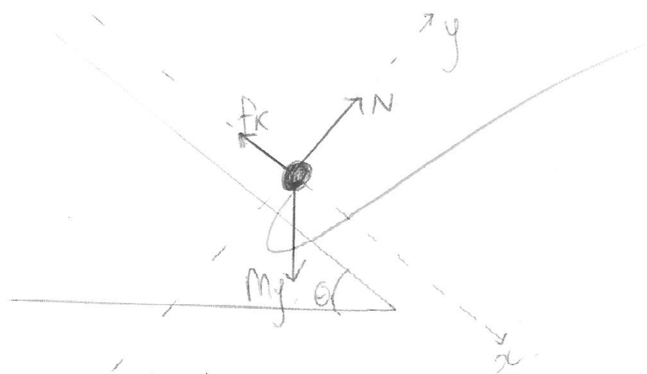
12

Question 4 / *Vraag 4* [12]

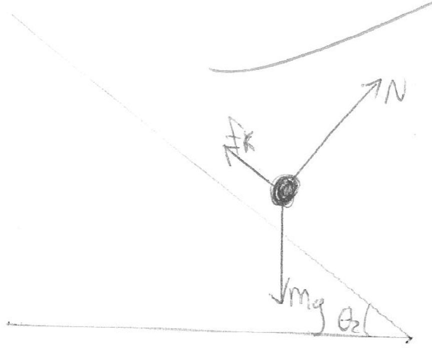
In a playground, two slides have different angles of incline θ_1 and θ_2 ($\theta_2 > \theta_1$). A child slides down the first at constant speed; on the second, his acceleration down the slide is a. Assume the coefficient of kinetic friction is the same for both slides. Find a in terms of θ_1, θ_2 and g.

In 'n speelparkie is daar twee glybane met verskillende hoeke ten op sigte van die horisontaal; θ_1 en θ_2 ($\theta_2 > \theta_1$). 'n Kind gly teen die eerste glybaan af teen 'n konstante spoed; op die tweede glybaan is sy versnelling a. Neem aan die koëffisiënt van kinetiese wrywing is dieselfde vir beide glybane. Vind a in terme van θ_1, θ_2 en g.

Child on slide 1, θ_1:



Child on slide 2, θ_2:



 ©University of Pretoria / ©*Universiteit van Pretoria*; 2016

12

x to the right as positive, y upwards is positive.

Slide 1:

x axis: ~~Mgsin θ=0~~

$\sum F_x = ma$

constant speed $a=0$ $\quad Mg\sin\theta_1 - f_K = 0$

$Mg\sin\theta_1 = \mu_K N_1 \quad -①$

y axis: $\sum F_y = ma$

$N_1 - mg\cos\theta_1 = 0$

$N_1 = mg\cos\theta_1 \quad -②$

③: $a = \frac{mg\sin\theta_2 - \mu_K N_2}{m}$

subs ④: $a = \frac{mg\sin\theta_2 - \mu_K mg\cos\theta_2}{m}$

$a = g(\sin\theta_2 - \mu_K\cos\theta_2)$

subs ⑤: $a = g(\sin\theta_2 - \tan\theta_1\cos\theta_2)$

Slide 2:

x axis: $\sum F_x = ma$

$Mg\sin\theta_2 - f_K = ma$

$a = \frac{Mg\sin\theta_2 - \mu_K N_2}{m} \quad -③$

y axis: $\sum F_y = ma$

$N_2 - mg\cos\theta_2 = 0$

$N_2 = mg\cos\theta_2 \quad -④$

①: $Mg\sin\theta_1 = \mu_K N_1$

subs ②: $Mg\sin\theta_1 = \mu_K mg\cos\theta_1$

$\mu_K = \tan\theta_1 \quad -⑤$

©University of Pretoria / © *Universiteit van Pretoria*; 2016

5

Question 5 / *Vraag 5* [6]

4

For an object that is moving on a circular path, and is subjected to a **centripetal** force, the centripetal acceleration is given by $a_c = \frac{v^2}{r}$, where v is the translational speed of the object, and r is the fixed radius of the circular path. By making use of the definitions of the **period**, T, and the **angular speed**, ω, prove that $a_c = r\omega^2$.

Vir 'n voorwerp wat op 'n sirkelbaan beweeg, en wat onderwerp word aan 'n ***sentripetale*** *krag, word die sentripetale versnelling gegee deur* $a_c = \frac{v^2}{r}$*, waar v die translasie spoed van die voorwerp is, en r is die vaste radius van die sirkelpad. Deur gebruik te maak van die definisies van die* ***periode****, T, en die* ***hoekspoed****, ω, bewys that* $a_c = r\omega^2$.

$a_c = \frac{v^2}{r}$ - (1).

$T = \frac{2\pi r}{v}$ (definition) - (2)

we can define: $\omega = \frac{2\pi}{T}$

subs - (2) $\omega = \frac{2\pi v}{2\pi r} = \frac{v}{r}$.

$\therefore v = r\omega$ - (3).

substitute (3) into (1): $a_c = \frac{(r\omega)^2}{r}$

$a_c = r\omega^2$

 ©University of Pretoria / ©*Universiteit van Pretoria*; 2016

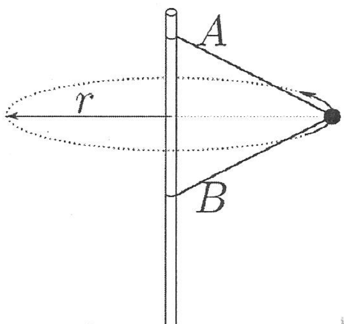
4

12

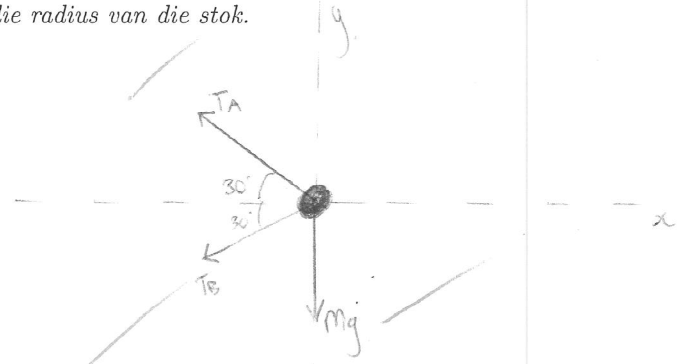
6 Question 6 / *Vraag 6* [12]

A child's toy has a 0.100 kg ball attached to two strings, A and B. The strings are also attached to a stick and the ball swings around the stick along a circular path in a horizontal plane. Both strings are 15.0 cm long and make an angle of 30.0° with respect to the horizontal. Draw the correct free-body diagram for the ball, and find the magnitude of the tension in each string when the ball's angular speed is 6.00π rad/s. The radius r is much larger than the radius of the stick.

'n Kinder-speelstuk het 'n 0.100 kg bal wat vasgemaak is aan twee toutjies, A en B. Die toutjies is ook vasgemaak aan 'n stok en die bal beweeg om die stok op 'n sirkelbaan in 'n horisontale vlak. Beide toutjies is 15.0 cm lank en maak 'n hoek van 30.0° met die horisontaal. Teken die korrekte vry-liggaam diagram vir die bal, en vind die grootte van die spankrag in elke toutjie wanneer die bal se hoeksnelheid 6.00π rad/s is. Die radius r is baie groter as die radius van die stok.



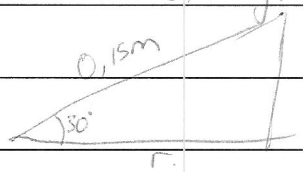
Ball:



x-axis: $\sum F_x = ma_c = m\frac{v^2}{r}$

$T_A\cos\theta + T_B\cos\theta = ma_c$ —(1)

m = 0,100kg



y-axis: $\sum F_y = 0$

$Mg - T_A\sin\theta + T_B\sin\theta = 0$.

$Mg + T_B\sin\theta = T_A\sin\theta$

$T_B = \frac{T_A\sin\theta - mg}{\sin\theta}$ —(2)

~~$\cos$~~

$\cos 30 = \frac{r}{0,15m}$

~~r = 0,023m~~

r = 0,13m

$\omega = 6,00\pi$

$v = r\omega$

$v = (0,13m)(6\pi)$

$v = 2,45$ m/s

$a_c = \frac{v^2}{r} = \frac{(2,45^2)}{(0,13)} = 46,17\,m/s^2$

Subs (2) into (1):

$T_A\cos\theta + \left(\frac{T_A\sin\theta - mg}{\sin\theta}\right)\cos\theta = ma_c$

$\frac{T_A\cos\theta\sin\theta}{\sin\theta} + \left(\frac{T_A\sin\theta - mg}{\sin\theta}\right)\cos\theta = \frac{ma_c\sin\theta}{\sin\theta}$

$T_A\cos\theta\sin\theta + (T_A\sin\theta - mg)\cos\theta = ma_c\sin\theta$

$T_A\cos\theta\sin\theta + T_A\sin\theta\cos\theta - mg\cos\theta = ma_c\sin\theta$

$2T_A\sin\theta\cos\theta = ma_c\sin\theta + mg\cos\theta$

$T_A = \frac{ma_c\sin\theta + Mg\cos\theta}{2\sin\theta\cos\theta}$

$T_A = \frac{(46,17\,m/s^2)(0,100kg)\sin 30° + (0,100kg)(9,8\,m/s^2)\cos 30°}{2\sin 30\cos 30}$

$= 3,65N$

TURN OVER.

 ©University of Pretoria / ©*Universiteit van Pretoria*; 2016

12

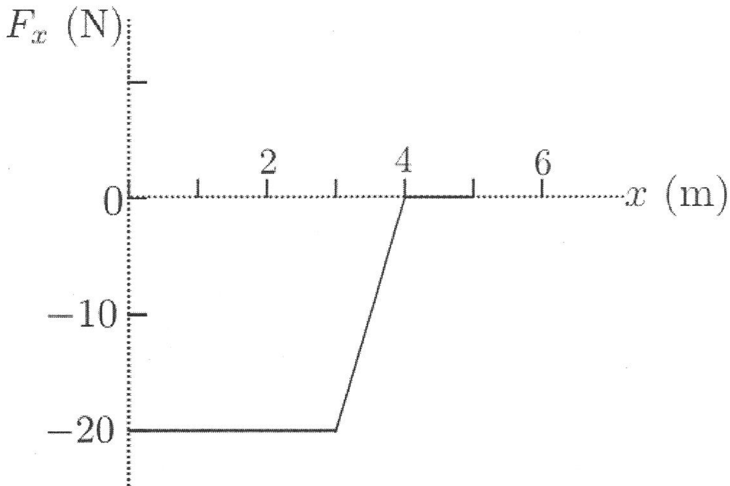
From (2): $T_B = \dfrac{T_A \sin\theta - mg}{\sin\theta}$

$= \dfrac{(3{,}65\,N)\sin 30 - (0{,}100\,kg)(9{,}8\,m/s^2)}{\sin 30}$

$= 1{,}69\ N$

Question 7 / *Vraag 7* [6]

A 100 g particle experiences the one-dimensional, **conservative force** F_x as shown in the figure.
*'n 100 g deeltjie ervaar die een-dimensionele, **konserwatiewe krag** F_x soos getoon in die figuur.*

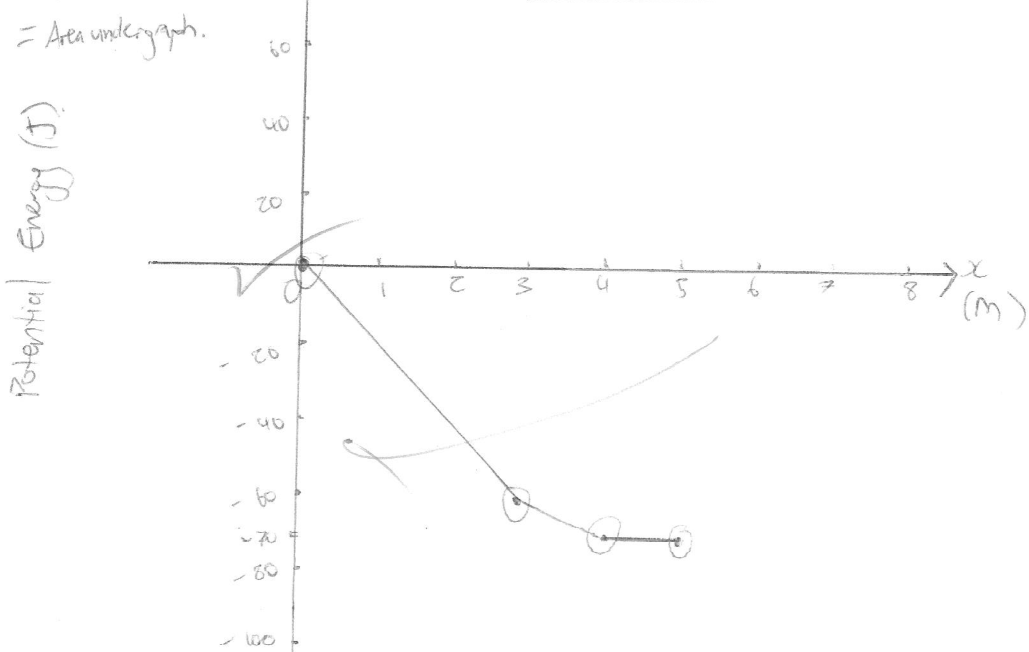


(a) Draw a graph of the potential energy U from $x = 0$ m to $x = 5$ m. Let the zero of the potential energy be at $x = 0$ m. **Hint: Remember the relationship between $F(x)$ and $U(x)$.** Make sure your graph is drawn according to scale and properly labelled.
*Teken 'n grafiek van die potensiële energie U vanaf $x = 0$ m tot by $x = 5$ m. Laat die die potensiële energie nul wees wanneer $x = 0$ m. **Wenk: Onthou die verwantskap tussen $F(x)$ en $U(x)$.** Maak seker dat jou grafiek volgens skaal geteken is en die regte benamings toon.* (4)

Graph showing potential energy of a 100g particle over 5m due to force F_x

$\Delta U = \int_{x_i}^{x_f} F_x \, dx$

= Area under graph.



 ©University of Pretoria / ©*Universiteit van Pretoria*; 2016

Displacement (m)

(b) The particle is shot toward the right from $x = 1$ m with a speed of 25 m/s. What is the particle's mechanical energy?
Die deeltjie word geskiet na regs vanaf $x = 1$ m met 'n spoed van 25 m/s. Wat is die deeltjie se meganiese energie? (2)

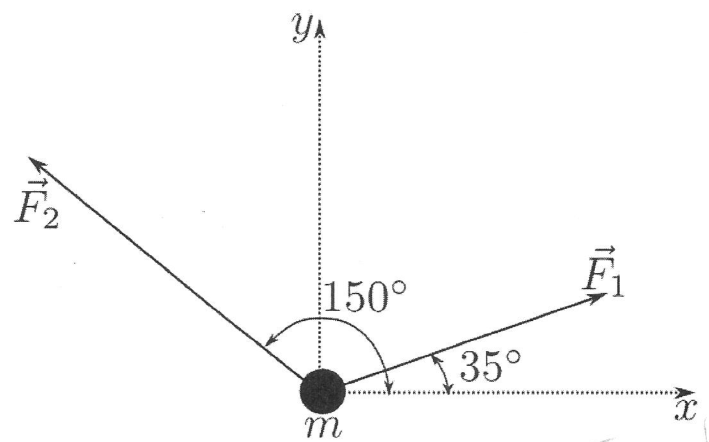
0

$E_{mech} = K + U$ U at $x=1$ is $\int_{3l_i}^{\infty} F\,dx$
$= \frac{1}{2}mv^2 + U_{x=1}$ $= -20$ J.
$= \frac{1}{2}(0{,}1\text{kg})(25\text{m/s}^2) + (-20\text{J})$
$= 11{,}25$ J.

Question 8 / *Vraag 8* [9]

5

Two constant forces act on an object of mass $m = 5$ kg moving in the xy-plane as shown in the figure. Force $\vec{F}_1$ is 25 N at 35°, and force $\vec{F}_2$ is 42 N at 150°. At time $t = 0$, the object is at the origin and has velocity $(4.0\hat{\imath} + 2.5\hat{\jmath})$ m/s. At the instant $t = 3.0$ s, use the work-kinetic energy theorem and find the final kinetic energy (at $t = 3.0$ s). Use the expression $\frac{1}{2}mv_i^2 + \sum \vec{F} \cdot \Delta\vec{r}$.
Twee konstante kragte werk in op 'n voorwerp met massa $m = 5$ kg wat in die xy-vlak beweeg soos getoon in die figuur. Krag $\vec{F}_1$ is 25 N teen 35°, en krag $\vec{F}_2$ is 42 N teen 150° gerrig. By $t = 0$, is die voorwerp by die oorsprong en dit het snelheid $(4.0\hat{\imath} + 2.5\hat{\jmath})$ m/s. Op die oomblik wat $t = 3.0$ s, maak gebruik van die arbeid-kinetiese energie teorie, en vind die finale kinetiese energie (by $t = 3.0$ s). Gebruik die uitdrukking $\frac{1}{2}mv_i^2 + \sum \vec{F} \cdot \Delta\vec{r}$.



$\bar{F}_1 = 25\text{N}(\cos 35°\,\hat{\imath} + \sin 35\,\hat{\jmath}) = (20{,}48\,\hat{\imath} + 14{,}34\,\hat{\jmath})$N $m = 5$kg
$\bar{F}_2 = 42\text{N}(\cos 150\,\hat{\imath} + \sin 150\,\hat{\jmath}) = (-36{,}37\,\hat{\imath} + 21\,\hat{\jmath})$N $V_i = (4{,}0\,\hat{\imath} + 2{,}5\,\hat{\jmath})$ m/s

We can calculate a using Newton 2:
$\bar{F}_{net} = m\bar{a}$
$\bar{F}_1 + \bar{F}_2 = m\bar{a}$
$(20{,}48\,\hat{\imath} + 14{,}34\,\hat{\jmath})\text{N} + (-36{,}37\,\hat{\imath} - 21\,\hat{\jmath})\text{N} = (5\text{kg})(a_x\hat{\imath} + a_y\hat{\jmath})\,\text{m/s}^2$
$\therefore \bar{a} = (-3{,}18\,\hat{\imath} - 1{,}33\,\hat{\jmath})\,\text{m/s}^2$

5

 ©University of Pretoria / © *Universiteit van Pretoria*; 2016

Calculate displacement:

$\bar{a} = (-3,18\hat{i} - 1,33\hat{j})\,m/s^2$

$\bar{v}_i = (4,0\hat{i} + 2,5\hat{j})\,m/s$

$\Delta t = 3s$

$\bar{r}_f = \bar{r}_i + \bar{V}_i t + \frac{1}{2}\bar{a}t^2$

For x: $\bar{r}_{fx} = \bar{r}_{ix} + \bar{V}_{ix} t + \frac{1}{2}\bar{a}_x t^2$

$= 0 + (4,0 m/s)(3) + \frac{1}{2}(-3,18 m/s^2)(3^2)$

$= -2,31\,\hat{i}\,m$

For y: $\bar{r}_{fy} = \bar{r}_{iy} + \bar{V}_{iy} t + \frac{1}{2}\bar{a}_y t^2$

$= 0 + (2,5 m/s)(3) + \frac{1}{2}(-1,33)(3^2)$

$= 1,52\,\hat{j}\,m$

Net force:

$\bar{F}_{net} = \bar{F}_1 + \bar{F}_2$

$= (20,48\hat{i} + 14,34\hat{j})N + (-36,37\hat{i} - 21\hat{j})N$

$F_{net} = (-15,89\hat{i} - 6,66\hat{j})N$

$K_f = \frac{1}{2}mv_i^2 + \sum \bar{F} \cdot \Delta\bar{r}$

$= \frac{1}{2}(5,0kg)(4,0\hat{i} + 2,5\hat{j})^2 m/s + (-15,89\hat{i} - 6,66\hat{j})N\,(-2,31\hat{i} + 1,52\hat{j})m$

$= 76,71\,J + 5,5\,J$

$= 82,21\,J$

END OF TEST / EINDE VAN TOETS

 ©University of Pretoria / ©*Universiteit van Pretoria*; 2016

0